

NERELLA SWATHI

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OBJECTIVE

ECE graduate aiming to secure a challenging role in the electronics field where I can utilize my skills in analog and digital electronics, circuit analysis, and problem-solving to support organizational growth.

EDUCATION

Sri Indu College of Engineering & Technology, Ibrahimpattanam	2021 – 2025
B. Tech – Electronics & Communication Engineering GPA: 7.4	
Sri Chaitanya Junior Kalasala, L.B. Nagar	2019 – 2021
Intermediate – MPC - 66%	
ZPHS - Chilpur	2019
SSC -90%	

PROJECTS

Sun Tracking Solar Panel System:

- **Automatic Sun Tracking:** The system continuously tracks the sun's movement from morning to evening, adjusting the solar panel's angle so it always faces the direction of maximum sunlight for better energy absorption.
- **Sensor & Control System:** LDR sensors detect light intensity from different directions, the microcontroller compares these values, and a motor is controlled to rotate the panel toward the brightest light source automatically.
- **High Efficiency Output:** By maintaining the optimal angle throughout the day, the system generates more electricity, improves overall efficiency, and performs better than traditional fixed solar panels.

Driver Drowsiness Detector:

- **Real-Time Monitoring & Detection:** The system uses IR sensors to continuously track the driver's eye movements and blinking patterns, while a microcontroller analyzes the data to detect signs of fatigue or prolonged eye closure.
- **Alert & Safety Mechanism:** When drowsiness is detected, the system instantly triggers a buzzer and LED alerts to warn the driver, helping prevent accidents and ensuring road safety.

INTERNSHIP

Data Analyst Internship Experience

July 2025

- Gained hands-on experience in data cleaning, EDA, and statistical analysis using **Excel, SQL, and Python** (Pandas, Matplotlib, Seaborn) to manipulate large datasets and build dashboard-ready visualizations.
- Developed and presented **data-driven business solutions**, translating complex findings into clear, stakeholder-friendly reports and visual storytelling derived from real-world datasets. Strengthened end-to-end analytical skills including **data wrangling, trend analysis, and KPI tracking**, aligned with industry-standard Data Analyst workflows and best practices.

SKILLS

Programming Language: C, Embedded C

Platform/Tools: Arduino IDE, MATLAB.

Hardware: Solar Panel, LDR Sensors, Microcontroller, Motor, Motor Driver, Power Supply, Buzzer, LED IR Sensors.

CERTIFICATIONS

Embedded Programming

Mastering Microcontroller and Embedded Driver Development

SQL